

Anna Yesypenko

Email: yesypenko.1@osu.edu Website: annayesy.github.io

Positions	Assistant Professor of Mathematics <i>The Ohio State University</i>	August 2025 - Current
	Postdoctoral Fellow <i>Oden Institute, University of Texas at Austin</i> Supervised by George Biros	Jan 2024 - August 2025
Education	University of Texas at Austin <i>PhD in Computational Science, Engineering, and Mathematics</i> <i>Msc in Computational Science, Engineering, and Mathematics</i> Advisor: Per-Gunnar Martinsson, Co-advisor: George Biros	Sep 2017 - Dec 2023
	Cornell University <i>Bachelor's of Science in Engineering, magna cum laude</i> Major: Computer Science	Sep 2013 - May 2017
Teaching	Math 6601: Numerical Methods in Scientific Computing I	Fall 2026
	Math 7611: Computational PDEs I	Fall 2025
Fellowships	CSEM Fellowship , The University of Texas at Austin.	Sep 2017 - Aug 2021
Preprints	[1] Simon Dirckx, Anna Yesypenko, and Per-Gunnar Martinsson. “A fast spectral overlapping domain decomposition method with discretization-independent conditioning bounds”. In: (2025). arXiv: 2510.25991 [math.NA]. [2] Joseph Kump, Anna Yesypenko, and Per-Gunnar Martinsson. “A Two-Level Direct Solver for the Hierarchical Poincare-Steklov Method”. In: (2025). arXiv: 2503.04033 [math.NA]. [3] Katherine J. Pearce, Anna Yesypenko, James Levitt, and Per-Gunnar Martinsson. “Randomized Rank-Structured Matrix Compression by Tagging”. In: (2025). arXiv: 2501.05528 [math.NA].	
Publications	[4] Anna Yesypenko and Per-Gunnar Martinsson. “Randomized Strong Recursive Skeletonization: Simultaneous Compression and LU Factorization of Hierarchical Matrices using Matrix-Vector Products”. In: <i>Journal of Scientific Computing</i> 106.3 (2026), p. 63. [5] Anna Yesypenko, Chao Chen, and Per-Gunnar Martinsson. “A simplified fast multipole method based on strong recursive skeletonization”. In: <i>Journal of Computational Physics</i> 524 (2025), p. 113707. [6] Anna Yesypenko and Per-Gunnar Martinsson. “SlabLU: a two-level sparse direct solver for elliptic PDEs”. In: <i>Advances in Computational Mathematics</i> 50.4 (2024), p. 90. [7] Anna Yesypenko and Per-Gunnar Martinsson. “GPU optimizations for the hierarchical Poincare-Steklov scheme”. In: <i>International Conference on Domain Decomposition Methods</i>. Springer. 2022, pp. 519–528. [8] Benjamin C Revard, William W Tipton, Anna Yesypenko, and Richard G Hennig. “Grand-canonical evolutionary algorithm for the prediction of two-dimensional materials”. In: <i>Physical Review B</i> 93.5 (2016), p. 054117.	
Thesis	[9] Anna Yesypenko. “Randomized algorithms for the efficient solution of elliptic PDEs on modern architectures”. PhD thesis. University of Texas at Austin, 2023.	

Invited Talks

CM+X Seminar at California Institute of Technology.

“Fast Randomized Recovery and Solvers for Rank-Structured Matrices”. 9 June 2026.

Seminar at University of Auckland.

“Randomized Compression and Factorization of Rank-Structured Matrices”. Auckland, New Zealand. 20 May 2026.

Seminar at Queensland University of Technology.

“Randomized Compression and Factorization of Rank-Structured Matrices”. Brisbane, Australia. 12 May 2026.

BIRS Women in Computational Methods for PDEs.

“Randomized Simultaneous Compression and Factorization for Hierarchical Matrices”. Banff, Alberta, Canada. 1-6 March 2026.

Invitations to Mathematics Seminar at The Ohio State University.

“Fast Randomized Solvers for Complex Physical Systems”. 15 October 2025.

Householder Symposium (plenary talk).

“Randomized Algorithms for the Simultaneous Compression and LU Factorization of Hierarchical Matrices”.

8-13 June 2025.

Babuška Forum at Oden Institute.

“Fast Randomized Solvers for Complex Physical Systems”. 18 April 2025.

SIAM Conference on Computational Science and Engineering.

“Advances in Domain Decomposition Methods and Fast Solvers” (organizer). 3-7 March 2025.

Seminar at The Ohio State University.

“Fast Randomized Solvers for Complex Physical Systems”. 30 January 2025.

Computational Tools for PDEs with Complicated Geometries and Interfaces at Flatiron Institute.

“An Introduction to the Hierarchical Poincare-Steklov Method”. 10-14 June 2024.

Copper Mountain Conference.

14-19 April 2024.

International Congress on Industrial and Applied Mathematics.

20-25 August 2023.

SIAM Annual Conference.

11-15 July 2022.

SIAM CSE Conference.

1-5 March 2021.

Workshops

Simulation-Based Optimization with Applications.

Providence, Rhode Island. 13-17 April 2026.

Randomized Numerical Linear Algebra.

Providence, Rhode Island. 2-6 February 2026.

Rising Stars in Computational and Data Sciences.

Austin, Texas. 12-13 April 2023.

Challenges in Computational Methods for Integral Equations.

Oaxaca, Mexico. 22-27 May 2022.

Gene Golub SIAM Summer School in Data Science.

Aussois, France. 17-28 June 2019.

Oberwolfach Seminar in Numerical Homogenization.

Oberwolfach, Germany. 09-15 June 2019.

Service and Outreach

Program Committee Member

Algorithms & Performance area/track for ISC High Performance 2026.

Mar 2026

Host of Babuška Forum

March 2021 - June 2022

The Babuška Forum series was started by Professor Ivo Babuška several years ago to expose students to interesting and curious topics relevant to computational engineering and science with technical content at the graduate student level.

Reviewer · National Science Foundation · Journal of Computational Physics · SIAM Journal of Matrix Analysis and Applications · Communications on Pure and Applied Mathematics · Applied and Computational Harmonic Analysis · Computers and Mathematics with Applications · International Journal of High Performance Computing Applications

Mentorship

Undergraduate research mentor

Raiyan Islam (Denison University, Mathematics, graduating Spring 2027; Five Colleges of Ohio) — *Efficient Randomized Data Processing with Statistical Guarantees*.

Chase Holm (The Ohio State University, Applied Mathematics, graduating Fall 2026) — *Fast Solvers for Elliptic PDEs and Boundary Integral Equations*.

Yun Qiao (The Ohio State University, Honors Mathematics, graduating Spring 2027) — *Structured Solvers for PDE-Constrained Optimization*.

Research mentor

Jan 2024 - Current

Joseph Kump in CSEM PhD program. Paper in review (listed above).

Directed Reading Program Mentor

Sep 2018 - May 2019

Directed Reading Program mentor at the University of Texas at Austin. Mentored undergraduate students in a computational research project.

Multiplying Polynomials via the FFT - Elizabeth Taber

Interpretable Matrix Decomposition - Emily Tallman

Software

- [10] Anna Yesypenko. *SkelFMM: A Simplified Kernel-Independent Fast Multipole Method in Python*. Version 1.0.0. Jan. 2025. DOI: 10.5281/zenodo.14613533. URL: <https://github.com/annayesy/skelFMM>.
- [11] Anna Yesypenko. *SlabLU: A Two-Level Sparse Direct Solver for Elliptic PDEs in Python*. Version 1.0.1. May 2024. DOI: 10.5281/zenodo.11238664. URL: <https://github.com/annayesy/slabLU>.
- [12] Anna Yesypenko. *SimpleOctree in Python*. Version v1.0.1. Jan. 2025. DOI: 10.5281/zenodo.14613403. URL: <https://github.com/annayesy/simpleoctree/>.